

memory-bench

A Screened Benchmark Dataset and Validity Study for
Organizational Memory in Agent Harnesses

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Abstract

An organization’s agents no longer run on one model: each person picks the harness built for their work. The weights organizational knowledge must reach are therefore plural and vendor-owned, so coherence cannot live in them. The memory layer is the only place every harness can reach. Of the ten published memory benchmarks we verified (§2), none measures whether a memory system holds that knowledge with the right scope, authority, and freshness.

We release memory-bench as an instrument for that question, and this paper is the evidence that the instrument measures what it claims. The dataset is three simulated software organizations: multi-week event streams delivered event-by-event to each principal who witnessed them; a hidden ledger from which the expected belief state $B(p, t)$ is computed rather than asserted; and behavioral work tasks injected at evaluation time rather than quiz questions. Every probe instance is paired with a counterfactual twin in which the probed fact differs and the rest of the stream is byte-identical. An instance is credited only when both sides pass. Across three seeds, 371 instances survive a five-gate screening pipeline under blinded judging; two further seeds are withheld unscreened as holdouts. All three released seeds miss the prespecified instance gate (125, 131, and 115 against 135). The misses are retained and marked rather than repaired.

The validity study is the result. A memoryless worker earns pair credit on 4 of 125 instances, with 95% upper bounds of 10.6%, 44.1% and 19.4% by capability rung. Probe validity is *task-model-relative*: under identical strict rules, 37/54 probe clusters survived the ceiling gate for one worker and 17/54 for a smaller sibling. Identical weights behind two agent harnesses agreed on only 65% of twin-ceiling outcomes, failing a prespecified 90% bar. Judge-human agreement on a blinded 150-pair packet reached 81.3% raw agreement at $\kappa = 0.537$, which fails our prespecified gate of $\kappa \geq 0.75$. Aggregate disagreement is symmetric, 14 items each way, but the symmetry is a cancellation: every judge error on absence-phrased criteria is an over-accept, and every error on scope criteria is an under-accept. Mid-study the provider deprecated the pinned worker. We report the event, release the partial rows as provenance, and specify the re-anchoring procedure that lets the dataset outlive any single worker. No comparative system result is claimed anywhere in this paper.

Keywords: agent memory; benchmark construction; construct validity; LLM-as-judge; counterfactual evaluation; organizational knowledge; reproducibility.

How to read this paper. This is an instrument and a validity study, not a leaderboard. Three claims: the released set measures organizational memory behavior at rungs 1–3 under a stated worker pin; pair credit via counterfactual twins filters priors; screening anchors are properties of a worker and a harness, not of the items alone. Four prespecified bars failed: the instance gate on all three released seeds, the cluster gate on seed 3, harness equivalence, and judge–human κ . Against them, one clean positive result: a memoryless worker earns pair credit on 4 of 125 instances (§4.5).

1 Introduction

1.1 Organizational knowledge has no home in the weights

Organizations are getting smaller while the number of agents working inside them grows, and the agents are not interchangeable. A developer’s coding agent, a designer’s creative tool, and a generalist assistant differ in system prompt, tools, and defaults, and each is chosen for those differences. This heterogeneity is a property of the ecosystem rather than a transitional untidiness to be standardized away.

One structural consequence follows. If the weights an organization runs on are plural and vendor-owned, then organizational coherence cannot live in them. The decisions, working rules, preferences, and commitments that make a company act like one company have to be held somewhere every harness can reach, and the memory layer is the only such place. On this view a memory system is not an accessory to an organization’s agents; it is the connective tissue.

The naive version of that idea does not work. Pooling every artifact and giving every agent the same view produces a rumor mill with perfect recall, not a collective intelligence. A personal preference is not org policy. A leadership decision outranks a loud opinion. A superseded plan must stop driving behavior while remaining retrievable. A need-to-know fact must not diffuse. What makes an organizational world model useful is that it is *scoped, tiered, and temporally correct*: each agent receives the context appropriate to its principal and its task, at the current epoch of truth.

1.2 What existing benchmarks measure instead

Published memory benchmarks overwhelmingly evaluate a single agent’s recall over a long conversation: can the system retrieve a fact it was told earlier. That is rung 0 of the capability ladder this benchmark is organized around (§3.3), it is table stakes, and long context saturates it. Organizational memory differs along three axes that single-agent recall does not exercise: events have multiple witnesses and per-principal visibility, facts carry tier and authority so that conflicts have correct rather than arbitrary resolutions, and facts are invalidated over time rather than merely accumulated.

1.3 What we claim, and what we do not

This paper claims an *instrument*, not a leaderboard.

We claim that the released dataset measures organizational memory behavior at rungs 1–3 of the ladder, for a two-team simulated software organization at L1–L2 scale, under a stated worker pin, and we report the measurements that support that claim alongside the ones that qualify it. This paper publishes no comparative system numbers. A benchmark that cannot say

what it measures is not a benchmark, so the measurements that qualify the claim are the claim. No result is ever reduced to a single aggregate score. Results are grouped by capability rung, always.

Three constraints were fixed before the evidence they govern was produced, and each is corroborated by dated git history rather than by assertion: the gate criteria and scoring rules, the worker and judge pins, and the commitment to carry failed gates as failures rather than repairing the probes that failed them. That commitment has a visible price. Four prespecified bars failed, and the items behind them are retained and marked rather than topped up (§4.1).

1.4 Contributions

1. **A screened dataset** (§3): 371 valid paired probe instances across three simulated organizations, with counterfactual twins, hidden ground-truth ledgers, and two unscreened holdout seeds. Every gate result is dated and corroborated by commit history.
2. **A construction and screening methodology** (§3): pair credit via counterfactual twins; per-item floor and ceiling validity screening with a pinned worker; blinded judging with a versioned rubric where the judge model is never the worker model nor the family that authored the criteria; a five-gate pipeline; and a statistics protocol prespecified before any result existed. The measurement problems this pipeline raises in its own right are the subject of a companion methodology paper [22].
3. **A validity study** (§4): the measurements that test whether the instrument works: floor validation, the task-model-relativity of probe validity, harness sensitivity, judge false-accept rate, and judge-human agreement.
4. **A durability procedure** (§4.6): what it takes to keep an agentic benchmark valid when the worker it was screened under disappears. We did not choose this contribution; the provider deprecated our pinned worker mid-study and we documented what that costs and how to recover from it.

1.5 What this paper does not contain

It contains no system comparison. The pilot that would have produced one was cut off when the pinned worker became unreachable (§4.6), and because probe validity is worker-relative (§4.2) a successor’s results cannot be attached to anchors measured under the old worker. The partial rows are released in §6 as provenance. It also contains no treatment of how a benchmark is validated when every component inside it is a language model; that is the companion paper’s subject [22], and the one result from it this paper leans on is cited in §4.4.

2 Related work

Citation note. These references were checked in two passes, because the section was first drafted from this project’s own field audit rather than from the papers. On 2026-09-14 identifiers and titles were resolved against their primary sources. All twenty arXiv identifiers resolve to a paper matching the name given, including the six that postdate the drafting model’s knowledge, and a deliberate nonexistent identifier was checked to confirm that failures report as failures. Three descriptions were wrong and are corrected here, and one claim was refuted by its own

source and rewritten rather than re-cited. Author lists were not in that pass’s scope. They were checked on 2026-09-17, which found seven entries carrying wrong given names and three titles that abbreviated what the source spells out; all ten are corrected in the bibliography below. That pass also read this section’s substantive claims about its sources against the sources themselves. MEMTRACK’s best model at 60% correctness and the three LoCoMo scores quoted from Mem0’s own Table 2 are as stated. One characterisation of BetterBench was not: it reports that most benchmarks give no statistical significance or uncertainty, which is weaker than calling that the criterion failed most often, and §2.4 now says the former.

2.1 What memory benchmarks currently measure

Table 1 places the ten benchmarks our audit covered. The established ones evaluate a single agent’s recall across a long conversation [1, 2, 4]. That is rung 0 of the ladder this benchmark is organized around (§3.3): recall on request, table stakes for a memory system, and saturated by long context. A second line moves toward agentic settings [7, 8, 9, 3], and MemoryAgentBench (2507.05257) [6] makes the argument that matters there, that pasting a conversation history into context evaluates memory off-policy, since a deployed system writes its memory incrementally rather than receiving the whole history at once. Every one of them is single-principal.

Two works are close enough that the distinction has to be drawn precisely rather than by category.

MEMTRACK (2510.01353) [5] is the nearest neighbour on the organizational axis, and it is not a conversational benchmark at all. It models realistic organizational workflows by interleaving asynchronous events across Slack, Linear, and Git, with noisy, conflicting, and cross-referring information, and it scores acquisition, selection, and conflict resolution rather than retrieval alone. Its best reported model reaches 60% correctness. Anyone reading our framing should read MEMTRACK first. What separates it from this work is not ambition but structure: MEMTRACK is single-agent, so it has no witness model and no notion of a fact reaching one person and not another. It scores conflict resolution without tiered authority deciding which side of a conflict is correct, and without supersession as a distinct failure mode from retrieval.

GateMem (2606.18829) [10] is the nearest neighbour on the multi-principal axis, evaluating memory governance for shared-memory agents across utility, access control, and deletion. It shares our premise that a memory system serving several people is a different object from one serving a single user. It scores access control on its own; we score governance jointly against propagation, because the two pull in opposite directions and a system can win either by sacrificing the other. That tension is the reason v1 defers leakage scoring rather than reporting it alone.

2.2 Why organizational memory is a different object

Single-agent recall does not exercise three properties that decide whether an organization’s agents behave coherently.

Events have witnesses. In a conversation benchmark every turn is visible to the one agent under test. In an organization a decision made in a meeting reaches the people in the room, and whether it reaches anyone else is the question being asked. Our streams are delivered per principal according to a witness model, so a system that pools everything and a system that respects visibility see different inputs by construction (§3.2).

benchmark	principals	what the system ingests	what is scored
LoCoMo (2402.17753)	one	conversation sessions	recall of a fact stated earlier
LongMemEval (2410.10813)	one	conversation sessions	recall, sometimes with a temporal qualifier
MemBench (2506.21605)	one	conversation sessions	recall, extending the same format
MemoryAgentBench (2507.05257)	one	memory written incrementally	agentic memory, measured on-policy
MemoryArena (2602.16313)	one	closed-loop agentic tasks	task outcome
StreamMemBench (2606.14571)	one	streaming egocentric lifelog	ingestion and feedback consolidation
HorizonBench (2604.17283)	one	preferences observed over time	preferences that change
LongMemEval V2 (2605.12493)	one	web-agent trajectories to 115M tokens	static and dynamic state, workflow knowledge, environment gotchas, premise awareness
MEMTRACK (2510.01353)	one	asynchronous Slack, Linear and Git events	acquisition, selection, conflict resolution
GateMem (2606.18829)	many	a shared memory store	utility, access control, deletion
memory-bench (this work)	many	a per-principal event stream, delivered event by event	scope, authority and freshness as behavior, credited per twin pair

Table 1: The ten memory benchmarks the 2026-07-25 audit covered, on the two axes that decide whether organizational memory is in range: how many principals hold different views of the same history, and whether that history arrives as a stream the system has to write down for itself. The first three are the rung-0 recall benchmarks. MEMTRACK is the nearest neighbour on setting and GateMem on governance (§2.5). Nine of the ten are single-principal, and the tenth holds one shared store rather than a per-principal stream.

Facts carry authority and tier. When a personal preference contradicts a team convention, or a loud opinion contradicts a leadership decision, there is a correct answer rather than an arbitrary one. Recall benchmarks have no notion of a fact outranking another fact, so they cannot score getting it wrong.

Facts are invalidated, not only accumulated. A superseded plan must stop driving behavior while staying retrievable on request. A benchmark that scores retrieval alone credits a system for surfacing a fact that should no longer apply.

Continual-learning and knowledge-editing work addresses the third property inside the weights. This benchmark takes the harness position instead: the knowledge an organization runs on changes faster than anyone retrains, and the agents consuming it run on models the organization does not own.

2.3 How memory evaluations have failed in practice

Most of this paper is validity evidence rather than results, and the reason is that this field’s published numbers have repeatedly failed on measurement rather than on modeling.

Ground truth that is wrong, and a judge that does not notice. An independent audit of LoCoMo [1] (Penfield Labs, 2026-04-08) found 99 score-corrupting errors in 1,540 questions, a ceiling near 93.6% that vendors were publishing above, and an LLM judge accepting topical

but wrong answers at 62.81%. It is self-published, and its author discloses the same finding in a public issue thread, so the figure has two routes rather than one.

A corpus that does not need memory. Mem0’s own evaluation (2504.19413, Table 2) [12] reports a full-context baseline at 72.90% against Mem0’s graph variant at 68.44% and plain Mem0 at 66.88%, and Letta separately reported a plain filesystem-and-grep agent at 74.0%. When the no-memory baseline wins on a vendor’s own table, the question is whether the corpus was measuring memory at all. A grep agent is one of our registered baselines for that reason: a memory product should have to beat a directory of files.

Protocol loose enough to support a dispute. One system was reported at 84%, then 75.14%, then 58.44% in a public exchange between Mem0 and Zep, depending on configuration and denominator, with neither party obviously acting in bad faith. MemDelta (2606.29914) [11] sharpens the point: an embedding-only swap moved accuracy 6.2 points ($p = 0.004$) while the architectural comparison it ran, verbatim RAG against full context, did not move (47.2 against 49.8, $p = 0.34$). Our RAG-class baselines pin and report their embedding model.

Fitting the system to the test after seeing it. A public teardown of MemPalace documents three patches hand-coded against three specific questions the system had failed, after which the result was reported as the first perfect score on LongMemEval [2]; the reporter’s reading is that the fixes were designed around the exact failure cases rather than found by analyzing general ones. The documented failure is test-set inspection followed by targeted repair, not benchmark content visible during ingestion, so what it argues for is a withheld ledger, regenerable holdout seeds, and a freeze that makes post-hoc repair visible in git history.

2.4 Benchmark validity and benchmark decay

The standards this paper is written against are BetterBench (2411.12990) [13], whose lifecycle assessment finds that most benchmarks report neither statistical significance nor uncertainty, 14 of the 24 it assessed; the Agentic Benchmark Checklist (2507.02825) [14], which requires auditing a scorer against degenerate strategies; the construct-validity audit of 445 benchmarks (2511.04703) [15]; Miller (2411.00640) [16] and 2503.01747 [17] on error bars and small-sample intervals; 2306.05685 [18] on measuring judge agreement rather than assuming it; and BenchBench (2407.13696) [19] on agreement testing between benchmarks.

The decay literature treats the *data* as the perishable part. GSM1k (2405.00332) [20] re-generated a held-out equivalent of a saturated benchmark and found accuracy drops up to 8% in some model families, and LiveBench (2406.19314) [21] answers rot with continuous refresh; both inform the withheld generator and unscreened seeds of §8. What none of it covers is the failure §4.6 reports. For an agentic benchmark the worker model used to establish which items are answerable is equally perishable, and it perishes on the provider’s schedule rather than on the benchmark’s. We measured how much validity depends on that choice (§4.2) and then lived through the dependency.

2.5 Where this benchmark sits

No benchmark in the audit scores tiered scope, authority weighting, propagation across principals, and supersession together as behavioral properties of an organization. The claim is a conjunction, and §2.1 says which neighbour breaks which part of it.

We state this as the result of a search rather than as a fact about the literature. The audit was compiled 2026-07-25 and names what it covered, the citations were verified against primary

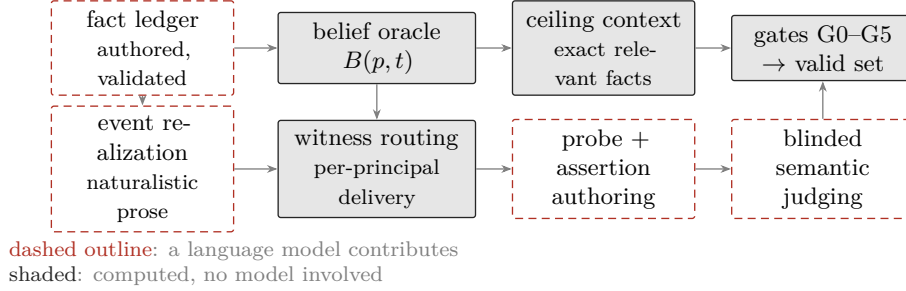


Figure 1: The construction pipeline. The path from the ledger to what is *true for a principal at a moment* runs entirely through shaded boxes: $B(p, t)$ is a pure function over the ledger and the event index. Models render events, author scoring criteria, and judge semantic criteria. No model decides what is true.

sources on 2026-09-14, and a reader who knows of prior work we missed should read the claim as bounded by that date and that list.

3 The dataset and how it was built

3.1 Ground truth is computed, not asserted

The design decision everything else rests on is that this benchmark’s answer key is a function, not a list of question–answer pairs written by a model (Figure 1).

Each simulated organization has a private ledger of facts: decisions, working rules, preferences, commitments, each with a tier (personal, team, project, org, external), a capacity (formal_decision > directive > opinion > speculation), a visibility, a validity window, and the events that evidence it. From the ledger and the event index we compute the **expected belief state** $B(p, t)$: every fact that principal is entitled to hold at that moment. A fact is in B when the principal witnessed it, when its visibility permits them, when it is valid and un superseded at t , and when its decay class still holds. Superseded facts move to a historical set B_{hist} rather than vanishing, because a superseded plan must stop driving behavior while staying retrievable.

Conflicts resolve by stated precedence: higher capacity first; at equal capacity the tier matching the *artifact* being produced wins, so an org policy governs an org-facing document and a personal preference governs a personal one; then recency. A pair that remains tied is not a defect to be broken arbitrarily but a genuine unresolved contradiction, where correct behavior is surfacing the conflict rather than silently picking. Rule two is what makes precedence contextual instead of a fixed hierarchy, and it is the mechanism the scope archetypes probe.

B is a pure function over (ledger, event index) and is the single scoring oracle. Ambiguity in B is a spec bug that blocks release rather than a judgement call resolved per item, and the two found this way were fixed before any screening.

3.2 Streams, witnesses, and what the system under test sees

Each organization’s history is a stream of naturalistic communication: meetings, DMs, team and org chat, email threads, documents, tickets, pull requests, calendar entries, with facts embedded in the prose as a person would express them, in the author’s voice, never in ledger-canonical form.

Fact annotations are stripped before the stream reaches a system under test; the annotated copy is private to scoring.

There is no global feed. The runner delivers each event only to the principals who witnessed it: participants always, plus anyone entitled to the surface it appeared on, since channel history is readable. A decision taken in a platform standup reaches a growth-team principal only if some later event carries it across, or if the memory layer under test does. That is what makes propagation a measurable property rather than an assumption, and it is why a system pooling everything and a system respecting visibility receive different inputs by construction. The adapter’s entire ingestion surface is one call, `ingest(principal_id, event)`, invoked once per witness per event; what a system does internally, whether one shared store or per-principal stores, is its own business, and it is scored against $B(p, t)$.

Realism is a set of generator obligations, not an aspiration. At least 60% of event content is operational filler embedding no probed fact; every probed fact carries at least two distractors, one of them a near-miss differing by a single attribute; no fact’s canonical text appears verbatim in more than one event, which blocks string-match shortcuts; canary strings sit in the text at low frequency for contamination detection; and timelines are sized so the annotation-stripped transcript exceeds roughly 500k tokens, which keeps full-transcript-in-context an honest but costly baseline rather than a free win.

Salience parity is achieved by construction and then tested. Distractor type, register, shape, length, and numeric form mirror the probed mix, and positions are assigned mechanically to class-stratified band centers rather than by re-rendering until a lint passes. A blinded discrimination check bounds it rather than confirming it: raters told probed excerpts from distractor-only ones at 58/100, 58% with a Wilson 95% CI of [48.2%, 67.2%] (binomial $p = 0.067$; protocol, per-seed splits and the human runs in the validation report). G2 set the bar at a reviewer who cannot beat 65% accuracy, and the interval’s upper bound reaches above that, so the check bounds discrimination rather than demonstrating flatness: the data do not exclude a discrimination rate at the level the gate itself calls failure. The comparison is approximate, because G2’s 65% is worded per 20-excerpt spot-check while 58/100 aggregates five of them. G2 was read off the point estimate, passed on that basis, and is not re-litigated here. The check does detect discrimination when it is large, which is how the one documented residual was found: seed 3 scored 14/20 on four independent samples, including after two targeted content revisions that failed to remove it (42/60 across the first three, 70% with a Wilson 95% CI of [57.5%, 80.1%], $p \approx 0.001$), and it ships disclosed, with users of salience-sensitive analyses pointed at the other seeds.

3.3 Probes are work, and every probe has a twin

A probe is not a question. It is a work task issued to a principal’s agent at a point in the timeline: write the announcement, draft the checklist, prepare the review note. What is scored is whether the resulting artifact behaves as the organization’s knowledge requires. Probes are injected only at evaluation time and never appear in the ingested stream, which is the countermeasure to tuning a system on the questions.

Four archetypes instantiate the capability ladder (Table 2). Rung 0, plain retention, is where existing memory benchmarks live and where long context saturates; it is implicit in every probe and never headlined. v1 is a deliberate vertical slice, at least one archetype on every rung above retention, rather than exhaustive coverage of any one rung.

Every probed fact has a counterfactual variant, and the generator produces a **twin orga-**

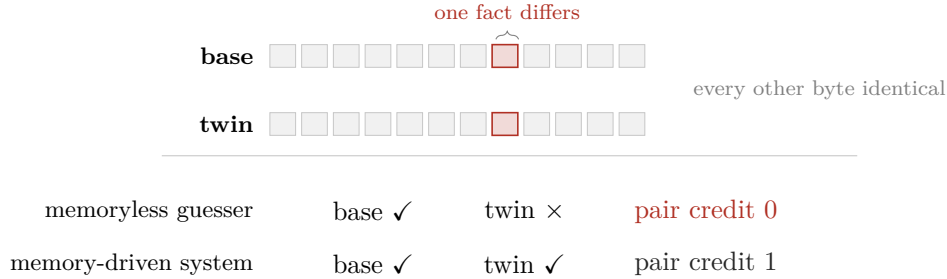


Figure 2: Pair credit via counterfactual twins. The twin organization is regenerated from the same seed with one fact changed and everything else held byte-identical. A guess that happens to match a model’s priors clears the base side and fails the twin, because the twin’s answer is the one those same priors argue against. Credit is the conjunction, so priors earn nothing.

rung	archetype	what the agent has to get right	instances
1, alignment	A4, tier collision	personal, team and org facts conflict, and context decides the winner	106
1, alignment	A7, supersession chain	a fact updated two or three times, probed at each epoch and once historically	172
2, coordination	A1, decision ripple	a decision taken in a meeting the probed principal did not attend	50
3, compounding	A2, silent rule	a working rule stated once, probed 30 or more events later with no reminder	43
valid paired instances, seeds 1 to 3			371

Table 2: The capability ladder as v1 instantiates it, and where the 371 valid paired instances sit on it. Rung 0, plain retention, has no row because it is implicit in every probe and never headlined. v1 is a vertical slice: no rung is covered by more than two archetypes, and the instance counts are what each rung’s result rests on.

nization from the same seed and the same timeline skeleton with the delta facts re-rendered (Figure 2). The same probe runs against both. An instance is credited **only when both sides pass**: correct behavior on the base organization, and correctly *different* behavior on the twin. Passing one side scores zero for the pair.

This is the most load-bearing design choice in the dataset. A fact invented by a language model often coincides with that model’s own prior, whether a naming convention, a review threshold, or a default cadence, and a memoryless system can guess the base side at a rate that would flatter it badly. It cannot guess both sides, and §4.5 reports what that buys.

Assertions come in several kinds, among them `fact_applied`, `fact_absent`, `scope_correct` and `conflict_flagged`. Each is checked by regex, by structural parse, or by a blinded semantic judge. Which checker is allowed where was settled by measurement rather than preference. Applied-content assertions written as regexes failed 39–42% of ceiling runs, because a deliverable paraphrases around any anchor; absence detectors failed 4% and semantic criteria 9–17%. Spec v0.3 therefore requires the semantic checker for applied content and reserves regex for absence detectors. Assertion text was model-authored under a fixed prompt and accepted only after mechanical validation, whose rules live with the spec rather than here. One authoring rule came out of failure adjudication and changed scoring, so it belongs in the paper: an assertion

must never punish content drawn from a different co-valid fact of the same cluster, because complementary facts are not competing answers.

3.4 Screening: which items actually measure memory

Every probe instance runs in three conditions. `floor` gives the task alone with no organizational context. `ceiling` gives the task plus exactly the relevant facts from $B(p, t)$, rendered canonically. `sut` gives whatever the system under test provides. A system’s score is normalized as $(\text{sut} - \text{floor}) / (\text{ceiling} - \text{floor})$.

The floor and ceiling conditions are the screen. An instance is valid when its ceiling passes, its twin ceiling passes, and its floor output fails at pair level. A ceiling that fails means the item measures reasoning rather than memory. The facts were supplied and the worker still could not produce the artifact. A floor that passes means the item is answerable without memory at all. Both get dropped, and the per-side floor pass rate survives as a reported `guessability` figure per cluster.

Two revisions to this rule were forced by measurement and dated before the evidence they govern. The floor gate moved to pair level, because canonical facts sometimes coincide with industry defaults and a per-side rule discarded valid pairs for the exact reason the twin design exists. And screening moved from clusters to instances: requiring all three instances of a cluster to be valid demands per-run reliability near 98%, since $0.9^6 \approx 0.53$ cluster survival follows even at 9% instance noise. A cluster now survives with at least 2 of 3 valid instances and ships only those, and both counts are reported throughout.

Semantic verdicts are produced blind. `judge-export` writes each output with its criteria under opaque ids, stripped of probe identity, condition, and base-versus-twin side; fresh judge agents read only that batch file; `judge-import` validates the ids and criterion keys before anything is recorded. The judge model is never the worker model and never the family that authored or repaired the criteria being judged, and the rubric is versioned verbatim in the repository. Blinding is enforced in code rather than by instruction. It is worth reporting that blinding changed little: an earlier unblinded pass over the same seed-1 material agreed with the blinded verdicts on 97.6% of criteria, 26 flips out of 1,092, split 14 True→False against 12 False→True with no systematic direction. Blinding is cheap insurance here rather than a large correction, and we would rather publish that null than imply we caught a bias we did not measure.

3.5 What the released dataset is

Seeds 1–3, probe-spec v0.4.3, judge rubric v2: **371 valid paired instances** (A1 50, A2 43, A4 106, A7 172). Per seed the valid-instance counts are 125/131/115, from cluster survival of 46/54, 46/54, 40/54; the stricter all-instances counts are 33, 39 and 35. Construction produced $54 \text{ clusters} \times 3 \text{ instances} \times 5 \text{ organizations}$, 810 instances in total, of which seeds 4 and 5 are withheld unscreened as holdouts.

The dataset was frozen on 2026-08-15 before any system was evaluated, corroborated by dated commits rather than by assertion.

3.6 Worker and harness pinning

All screening ran under a single pinned worker: gpt-5.4 at medium reasoning effort behind codex-cli 0.144.5, with the binary pin enforced in code rather than by convention, because the system

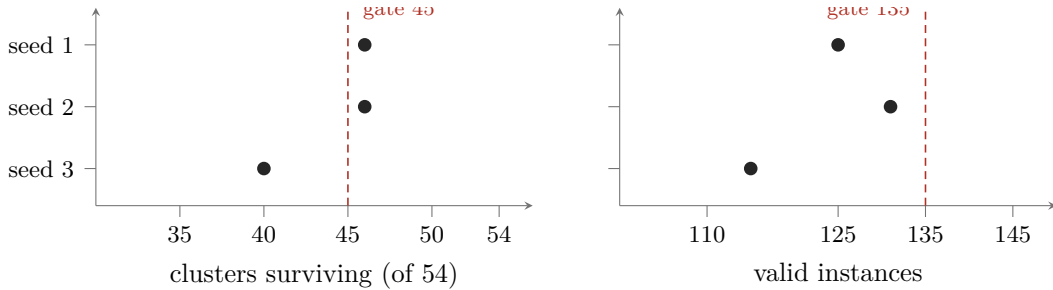


Figure 3: Gate G3 by seed. Every point left of a dashed rule is a failure carried into the release rather than repaired. All three instance gates fail; seed 3’s cluster gate fails as well. Nothing was topped up to cross a line.

codex upgrades silently. The pin is not housekeeping: §4.2, §4.3 and §4.6 are each a consequence of it.

4 Validity study: the results core

Every number below was already measured; none of it needed the deprecated worker.

4.1 Gate results

Six gates run during construction, and their outcomes are reported as measured. G0 and G1, the belief-oracle test suite and the schema, oracle cross-check, ambiguity scan, and interleaving checks, passed on every seed. G2’s machine side passed: structural regeneration, realization checks, and the release-blocking stream lint covering marker round-trip, visibility consistency, salience permutation tests, a noise floor, and canaries all report green across five organizations and their twins. Two of G2’s human items remain open by the freeze, and the salience residual on seed 3 is documented rather than repaired (§3.2).

G3 is where the failures are (Figure 3), and they are the reason the rest of this section is worth reading. **All three instance gates fail** against a prespecified 135, and **seed 3’s cluster gate also fails** at 40 surviving clusters against a floor of 45.

Table 3: Gate G3 by seed, carried as measured. Every **FAIL** is a shortfall released rather than repaired: no failing probe was rewritten and no threshold moved once it bound.

seed	clusters surviving	valid instances	instance gate (≥ 135)	cluster gate (≥ 45)
1	46/54	125	FAIL	pass
2	46/54	131	FAIL	pass
3	40/54	115	FAIL	FAIL

The shortfall is structured rather than random. It is the residual tail of counterfactual-anchoring failure on the twin side, the same defect family that drives cluster drops, and the prespecified v2 response is to over-generate four instances per cluster and require counterfactuals to invert the aspect the task actually elicits.

Two further prespecified bars failed: the harness-equivalence study of §4.3, and G4 in §4.4. Four failures against one clean positive result, the memoryless floor of §4.5.

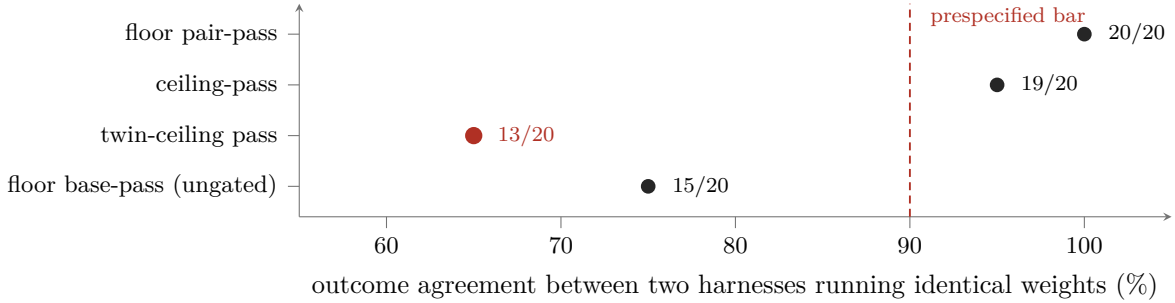


Figure 4: Harness sensitivity. Sixty runs, 20 per condition, identical gpt-5.4 weights, identical prompts, identical blinded scoring; only the agent scaffold differs. Agreement is near-total where the task is easy to fail or easy to pass, and collapses in the one condition that is hard to satisfy. The study failed its own acceptance rule, so the second harness was confined to non-protocol work.

No probe that failed a gate was repaired to recover it, no criterion was rewritten after seeing that it cost an instance, and no threshold moved once it bound. Where repairs did happen they were made blind to direction and before the evidence they touched: the cross-organization lineage sweep ran without knowing whether it would move counts up or down, and it moved them both ways.

A gate that is relaxed once it binds was never a gate, so we would rather publish a dataset that fails its own thresholds than one whose thresholds were chosen after the fact. That is the whole argument for trusting the numbers that did pass.

4.2 Probe validity is task-model-relative

Screening asks whether a worker given exactly the right facts can produce the artifact the task requires. That question has a different answer for different workers, and the size of the difference is the finding.

Under identical strict rules, **37 of 54** probe clusters survived the ceiling gate for gpt-5.4 and **17 of 54** for gpt-5.4-mini, a smaller model from the same family. Less than half. Nothing about the probes changed between those two numbers; the items are the same items and the rule is the same rule.

The consequence is that a valid item is not a property of the item. It is a property of the pair, item and worker, and a released valid set is defined relative to the model that screened it. This is not a quirk of our design. Any benchmark that decides which of its items are answerable by checking whether a model can answer them inherits the same relativity, whether or not it says so. Most do not say so, and many do not fix the worker at all, which makes their item sets quietly unreproducible.

We handle it by pinning the worker, reporting the pin everywhere a number appears, and publishing the per-worker re-screening procedure so a future user can re-derive a valid set under whatever model they have. Section 4.6 is what happens when that future arrives sooner than expected.

4.3 Harness sensitivity

Before admitting a second agent harness to any protocol path we ran a prespecified equivalence study: 60 seed-1 runs, 20 per condition, RNG seed fixed, scored through the identical blinded

pipeline, with the acceptance rule written down before any run. The bar was outcome agreement $\geq 90\%$ per condition. The two harnesses ran identical gpt-5.4 weights at identical effort; only the agent scaffold differed.

The study failed its own acceptance rule (Figure 4), and the second harness was confined to QA and non-protocol work on that evidence.

The disagreement is not uniform. It concentrates in the conditions that are hardest to satisfy. Where the task is easy to fail (the memoryless floor, at 100%) or easy to pass (the ceiling with facts injected, at 95%) the harnesses agree. The twin ceiling asks a worker to produce a deliverable consistent with a counterfactual fact while an almost identical base fact is absent, and there the same weights behind different scaffolds disagree on more than a third of outcomes.

One consequence generalizes past this study. The harness is part of the consumer, not a neutral pipe to the weights. A benchmark that pins a model and not its harness has not pinned its worker, and a result reported without a harness version is unanchored. It also explains why re-anchoring (§4.6) has to fix the harness version alongside the model.

4.4 Judge validation

We ran two checks on the judge: a cheap automated one and an expensive human one. They disagree about whether the judge is sound, and the reason they disagree is the most transferable result in this paper.

The 20-decoy false-accept audit: PASS, and blind to the defect

Deliberately wrong-but-topical outputs were judged blind against the real criteria: **2/41 false accepts as measured, 0/39 after adjudicating two criteria an under-specified decoy legitimately satisfied**. Read on its own this says the judge does not accept plausible-sounding wrong answers, which is the LoCoMo failure mode of §2.3.

Resolving those 41 criteria back to their kinds shows why that reading was premature: **30 were fact_applied, 11 were scope_correct, and none were fact_absent**. A decoy is built by stating wrong values, so it exercises criteria that demand content. A criterion phrased as absence is satisfied by an output that never raises the topic, so a decoy cannot probe it without being built differently. The audit was structurally incapable of detecting a failure on absence criteria, and that is exactly where the failure turned out to be.

G4 judge-human agreement: measured 2026-09-14, FAIL

The blinded 150-pair packet was rated by the single author-rater (the disclosed downgrade, §7 item 2) and scored against the committed judge verdicts.

The intervals are a percentile bootstrap over the 89 fact clusters the packet draws on rather than over the 150 items, because the packet is stratified with a per-cluster cap: 10,000 replicates, fixed seed, zero degenerate replicates. They were computed on 2026-09-17, after the gate was measured, so they are post-hoc and were never prespecified. G4 was prespecified and decided on the point estimate. The upper bound of the overall interval is 0.688, below the gate, so the failure does not depend on the point estimate. BetterBench [13] finds that most benchmarks report no uncertainty on their results at all, and putting an interval on our own headline statistic is the standard we cited it for. The per-kind intervals overlap heavily and establish nothing about

Table 4: G4 judge–human agreement, overall and by criterion kind. The gate fails on the point estimate and on the upper bound of the interval, so the verdict does not turn on either. `fact_absent` is where agreement collapses; the per-kind intervals overlap too heavily to rank the kinds against each other.

	n	raw agreement	κ	bootstrap 95% CI
overall	150	0.813	0.537	[0.370, 0.688] (gate $\kappa \geq 0.75$: FAIL)
<code>fact_applied</code>	72	0.819	0.605	[0.388, 0.782]
<code>scope_correct</code>	34	0.794	0.561	[0.298, 0.837]
<code>fact_absent</code>	44	0.818	0.166	[0.000, 0.493]

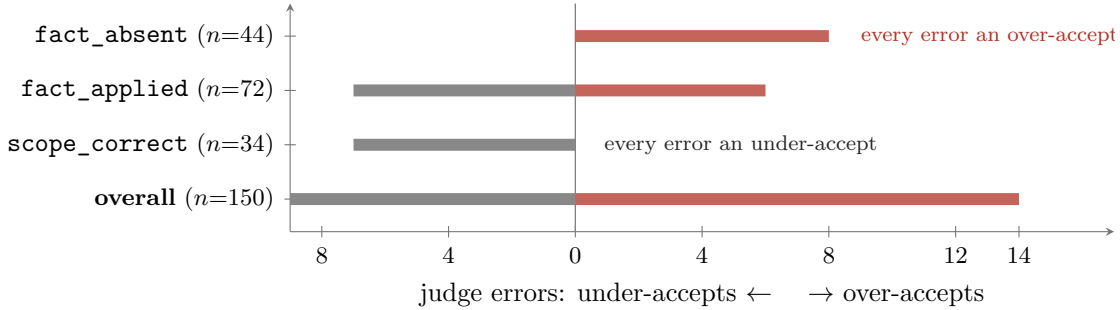


Figure 5: The symmetry is a cancellation, not a property. On absence-phrased criteria every judge error is an over-accept; on scope criteria every error is an under-accept. They cancel to 14 against 14 in aggregate, and the bias index is exactly 0.000, purely because this packet happens to contain 44 of the first kind and 34 of the second. A different mix of criterion kinds would not cancel.

whether κ differs by criterion kind. The kind-wise claim below rests on the one-sided error direction and on the corpus-scale positive rates, not on these intervals.

The headline number alone misleads in both directions.

The symmetry is a cancellation, not a property. Aggregate disagreement splits 14 and 14, and both raters label 72.0% of items positive, which invites the conclusion that the judge is unbiased noise. The per-kind confusion matrix says otherwise (Figure 5): on `fact_absent` the judge over-accepts on all 8 of its errors, on `scope_correct` it under-accepts on all 7, and `fact_applied` is genuinely mixed at 7 and 6. Any system scored on this benchmark inherits a bias whose sign depends on which archetypes its instances draw from, which is a property of the judge that no single agreement number reveals.

The judge barely discriminates on absence criteria. It returns FALSE on 1 of 44 absence-phrased items; the human rater returns FALSE on 9. A criterion of the form “does not present X as current” is, in this judge’s hands, close to automatically satisfied.

$\kappa = 0.166$ on `fact_absent` is a base-rate artifact, not evidence that absence criteria are harder to agree on. Raw agreement on that kind is 0.818, indistinguishable from `fact_applied` at 0.819 and `scope_correct` at 0.794; κ collapses only because the judge’s 97.7% positive rate pushes chance agreement to 0.782. We say so explicitly because the opposite reading is the natural one and we made it ourselves in an earlier draft.

The gate was $\kappa \geq 0.75$. It failed at 0.537, and that stands. One diagnostic is worth a sentence, because the opposite reading is the tempting one: the collapse on `fact_absent` is the kappa paradox, where a dominant label pushes chance agreement up and κ then penalizes the residual disagreement out of proportion to its size. Prevalence-adjusted statistics, computed

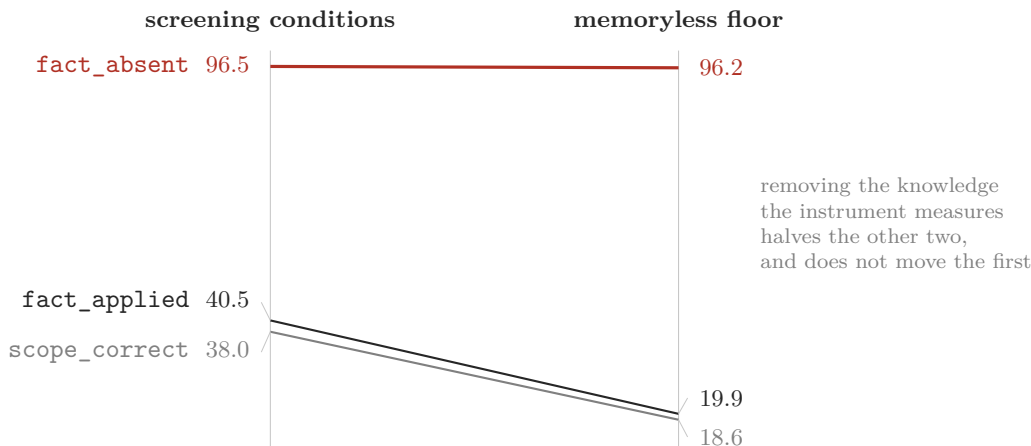


Figure 6: Judge positive rate per criterion kind (%), screening conditions against the memoryless floor. Under a worker with no organizational context, **fact_applied** and **scope_correct** roughly halve, which is what a discriminating criterion does. Measured on all 5,398 committed rubric-v2 verdicts, with no human labels involved. **fact_absent** does not move: a criterion class that ignores the removal of the knowledge under test is passing by construction.

post-hoc and never prespecified, put the same data between 0.59 and 0.77. We do not offer that as a defence. We prespecified κ rather than raw agreement precisely so an unbalanced task could not be dressed up as validity, and we do not get to discover the objection to our own gate on the day it fails.

Where the leniency lands

The over-accepts concentrate on the counterfactual side. Of the judge’s 14 over-accepts, 11 fall on twin-side criteria against 3 on base-side; its under-accepts split evenly, 7 and 7. This follows from the design. A twin asserts that the base fact is *not* presented, so absence-phrased criteria live disproportionately on that side (26 of the 44 sampled absence items). Pair credit requires both sides to pass, so a judge that waves twin sides through inflates instance credit directly. The error is in the direction that flatters a system under test.

The exposed fraction of the dataset is not small. 218 of the 371 valid instances (58.8%) carry at least one **fact_absent** criterion: 199 on the base side, 207 on the twin side, 188 on both.

The mechanism is not mysterious. “The note does not present the 48h SLA as current” is satisfied by a note that never mentions the SLA at all, so silence passes. This is therefore as much a criterion-design defect as a judge defect.

The mechanism at corpus scale, without human labels

The mechanism, a criterion class with a degenerate pass mode, can be measured without human labels at all, on every verdict the project has committed.

Joining all **5,398** committed rubric-v2 criterion verdicts to their assertion kind gives the judge’s positive rate per class (the join is deterministic and lossless): **fact_absent 96.4%** ($n = 1,463$), **fact_applied 39.0%** ($n = 2,677$), **scope_correct 36.9%** ($n = 1,258$). The 97.7% observed

on 44 sampled items was not a sampling artifact. The rate holds across all five sources of verdicts with no exception, from 93.2% to 98.9%.

The memoryless floor run supplies the control that makes this diagnostic rather than merely descriptive (Figure 6). Under a worker that demonstrably cannot know the facts, `fact_applied` collapses to 19.9% and `scope_correct` to 18.6%. `fact_absent` holds at **96.2%**, statistically unmoved. A criterion class that does not respond to removing the very knowledge the instrument is built to measure is not discriminating; it is passing by construction, and the two criterion kinds beside it in the same runs, judged by the same judge under the same rubric, show what responding looks like.

This is a re-analysis of already-committed verdicts, prespecified and dated before it was computed: no output was re-run, no criterion re-judged, and no valid set moved. It is a statement about the mechanism, never about accuracy.

What the defect does not reach

The leniency is contained by a structural property of the probe set, which we checked rather than assumed: **no side of any instance is scored on absence criteria alone**. Every base side and every twin side across all 371 valid instances pairs its absence criteria with at least one criterion demanding positive content. An output that stays silent, or that knows nothing, cannot pass even one side on the judge’s leniency.

The floor run bears this out empirically. Under a memoryless worker, 4 of 125 instances earned pair credit, which is the result reported in §4.5. If twin-side leniency were letting ignorance through, the floor would sit well above 4 of 125 and the rung-1 upper bound would not stop at 10.6%. **The judge defect therefore does not explain away this paper’s one positive result.**

What it does affect is the size of any future number on absence-heavy archetypes. Exposure is uneven: A4 70.8% of valid instances, A7 57.0%, A2 55.8%, A1 42.0%. Any score later computed on A4 in particular should be read as an upper bound rather than an estimate.

Why we did not fix it, and why fixing it would not have passed

Recomputing κ on the same 150 pairs with the identified defect removed is assumption-free, because it only requires flipping verdicts we already have. **Repairing the defect we found, perfectly, still fails the gate:** removing all 8 absence false-accepts takes agreement to 0.867 and κ to 0.688. The residual is symmetric noise on `fact_applied` plus 7 false-rejects and zero false-accepts on `scope_correct`, where the judge is already too strict, and rubric strictness is one dial, so turning it up to fix absence criteria makes scope criteria worse. Fixing both biases in opposite directions lands at 0.787, a hair over the line and assuming a precision no rubric edit delivers. The G4 failure is therefore not attributable to the absence defect alone.

Even if the numbers worked, the criteria were frozen before this evidence existed, and rewriting them now is the practice §2.3 condemns. The freeze that forbids this repair is what makes every other number here checkable against dated git history, and it cannot be spent selectively on the gates that fail. The subtler trap is that iterating rubric versions until κ clears 0.75 stops measuring judge quality and starts measuring how many attempts were taken, and an unbiased re-measurement needs a held-out calibration set we do not have.

Repair also has a price in artifacts. All 5,398 criterion verdicts committed under rubric v2 would need re-judging, since a v3 verdict cannot be mixed with a v2 one. That is mechanically

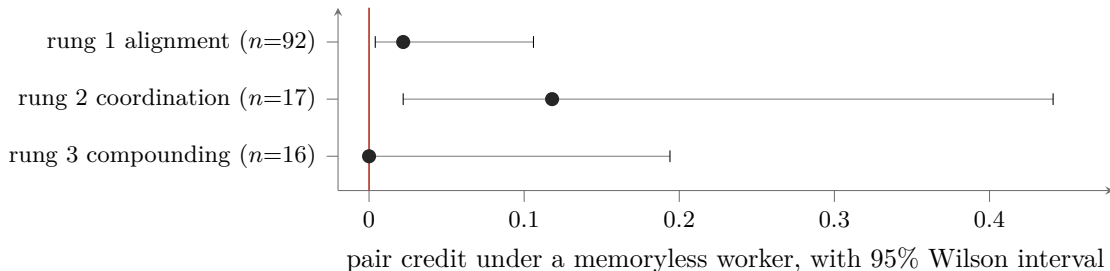


Figure 7: Floor validation. A worker with no organizational context earns pair credit on 4 of 125 instances: 2 of 92, 2 of 17 and 0 of 16 by rung. Bars are Wilson score intervals on the cluster-adjusted effective sample size, so they are asymmetric and never run below zero. Only rung 3, at 0 successes, is consistent with exactly zero; rung 2 rests on 17 instances and its interval reaches 44.1%.

possible without the deprecated worker, because worker outputs are cached and re-judging is judge-side, but the 371-instance valid set is *defined* by rubric-v2 verdicts, so a stricter judge moves the dataset itself. That is a v2 project with a new rubric and a fresh held-out calibration packet, not a patch.

The prespecified v2 fix: pair every absence criterion with a positive criterion demanding the replacement value, so the class stops having a degenerate pass mode.

One honest limit on the diagnosis, now narrower than it was. The two halves of the claim rest on different evidence. That the class has a **degenerate pass mode** is demonstrated without human labels, on 1,463 verdicts, with the floor run as a control. That the judge’s permissiveness is **wrong on particular items** still rests on one rater, and with one rater we cannot separate judge error from rater error. A second independent rater would settle the second half and would not change the first.

A judge panel is not that rater. The companion methodology paper [22] had four further blinded judges re-judge this packet’s semantic criteria. They agree with each other far above the gate, κ 0.927 to 0.966 against the committed v1 judge across 478 criteria, and with the human rater far below it, κ 0.518 to 0.563 on the 150 sampled pairs. On the 44 sampled absence criteria all four returned identical verdict vectors. Majority, unanimous-AND and unanimous-OR aggregation land at κ 0.537, 0.524 and 0.544, at or below the best single judge at 0.563. Those are that paper’s numbers, and the consequence for this one is that more judges do not substitute for the second human rater of disclosure 2, because the judges’ failure on that criterion class is perfectly correlated.

The methodological point, which generalizes past this benchmark

Two judge-validity checks, run on the same judge under the same rubric, returned opposite verdicts. The cheap automated one passed and could not have failed, because decoys are built by stating wrong values and therefore exercise only criteria that demand content. The expensive human one failed and localized a specific degenerate class. Anyone building a benchmark judged by a language model should report judge agreement **per criterion type**: an aggregate false-accept rate, and even an aggregate over/under-accept balance, can conceal opposite directional failures that cancel.

4.5 Floor validation: pair credit filters priors

This is the paper’s one clean positive result, and it tests the claim the whole design rests on: that the instrument measures memory rather than competence.

The memoryless condition ran end to end through the full evaluation pipeline on seed 1, not as a special case. A worker with no organizational context received each of the 125 valid instances on both the base and twin sides, and the 250 resulting deliverables were judged blind through the same export and import path as everything else.

Pair credit is small on every capability rung, and the counts are what carry it (Figure 7): **rung 1 pair credit 0.022**, 2 successes of 92, 95% Wilson interval [0.004, 0.106], $n = 92$, for alignment; **rung 2 0.118**, 2 of 17, [0.022, 0.441], $n = 17$, for coordination; and **rung 3 0.000**, 0 of 16, [0.000, 0.194], $n = 16$, for compounding. Four instances out of 125 earned credit in total. Only rung 3, with no successes at all, is consistent with exactly zero, and rung 2’s interval reaches 44.1% on 17 instances, which constrains almost nothing.

The interval method changed on 2026-09-17. Earlier drafts reported the Wald normal approximation, which is invalid at these counts and degenerates to $[0, 0]$ at rung 3’s zero successes, the failure 2503.01747 [17] warns against and the reason this paper cites it. These are Wilson score intervals on the cluster-adjusted effective sample size $n_{\text{eff}} = n/d_{\text{eff}}$, where the design effect d_{eff} is 1.80, 1.88 and 1.00 by rung. This is a reporting correction and nothing more. No valid set moved.

The mechanism is the twin (§3.3). A memoryless worker can guess a base side at a rate well above zero, because a generated fact often matches the priors of a competent model, and a benchmark crediting single sides would read that as memory. The design predicted this and the floor measures it.

Two honest qualifications. Rung 2 rests on 17 instances and rung 3 on 16, so the intervals are wide and the rung-level claim is weak even though the direction is not. And this is one seed, because the deprecation stopped the other two before they ran. What the result supports is a magnitude rather than a null test: a memoryless worker earns pair credit on 4 of 125 instances, with 95% upper bounds of 10.6%, 44.1% and 19.4% by rung. That is what the claim that the instrument does not reward priors or generic competence rests on, and no more.

4.6 Benchmark durability: the deprecation event and re-anchoring

The dataset was frozen on 2026-08-15. Pilot runs began 2026-09-02. On 2026-09-04 the runs stalled, and what the logs recorded was not an announcement but a wall of identical worker failures. By the time it was diagnosed the provider had removed gpt-5.4 from every billing path available to this project, returning a hard error. The pinned worker had ceased to exist, roughly seven weeks after being pinned.

Section 4.2 makes the consequence unavoidable. Screening anchors are properties of a worker, so if the worker is gone the anchors cannot be reproduced, the partial rows cannot be completed, and rows collected under a successor cannot be mixed with rows collected under the dead one. The decay literature of §2.4 answers a perishable dataset with regeneration and refresh; for an agentic benchmark the data is the durable part and the worker is the perishable one. We would not have written that up as a contribution if it had not happened to us.

The release therefore ships a maintenance contract rather than an assurance. If a successor worker is pinned, the rule is the nearest same-provider successor at the time of deprecation; anchors are re-screened for the released seeds under the frozen gate rules with task text held

byte-identical, which is what keeps the comparison a re-anchoring rather than a new dataset; the result is a new valid set under the new worker, reported as such; and only then can systems be evaluated. Anchors and system rows from different workers are never mixed. The procedure is specified whether or not we ever run it, because a dataset whose validity cannot be re-established outlives nothing.

5 The instrument as released

The dataset ships with the harness that runs it, because a benchmark whose evaluation code is a description rather than an artifact cannot be reproduced.

The adapter surface is three methods. A counters dictionary, `ingest(principal, event)`, and `run_task(principal, task) -> str`. Memory internals are never inspected. What a system stores, how it indexes, when it consolidates: none of it is observed or scored, only the behavior of the artifact its worker produces.

Witness routing lives in the runner rather than in any adapter, so per-principal delivery is identical across systems and an unrecognized visibility value raises rather than defaulting (§3.2). A worker failure is recorded as a null deliverable carrying its error, never as a dropped run, so a system cannot improve its score by failing.

Eleven system configurations are registered; ten are runnable in v1. Four baselines (the memoryless floor, full-transcript long context, a filesystem-and-grep agent, embedding RAG), a lexical BM25 ablation of the RAG baseline, four market systems selected by published inclusion criteria, and a per-principal silo ablation of one of them. The eleventh, a typed-memory reference implementation, is registered but deferred by the v1 freeze and never run. The grep baseline is there on principle (§2.3).

Configurations were frozen before any live run and amended only mechanically. The per-system config file was committed 2026-08-27, before any system was contacted, and states that the only permitted post-smoke changes are availability fixes, meaning auth flags, timeouts and API-shape corrections, recorded as dated amendments, never retrieval-quality tuning. Four amendments followed on 2026-09-02, all dated in git.

Two of the four are worth naming, because a reader could contest whether they are purely mechanical: one vendor’s search was pinned to the hybrid mode its own documentation recommends after extracted-memories-only searches returned empty, and another’s models were pinned after the configured defaults returned 404. Both were forced by a failing result rather than chosen to improve ranking, and both are still configuration changes, so we say so. The other two namespace container tags per organization side and handle null timestamps.

One embedding model sits behind every RAG-class baseline, `gemini-embedding-001`, pinned in code with the pin asserted by tests, and a retriever constructed without one identifies itself as `UNPINNED` in the artifacts it writes, for the reason §2.3 measures.

The scoring pipeline derives every number from artifacts. Scoring joins the screening anchors, the frozen valid set, and the SUT rows into one manifest, hard-failing on a missing base or twin output rather than scoring a partial pair. Aggregates carry cluster-robust standard errors computed with a design effect from an estimated intra-cluster correlation, because probe instances cluster within fact clusters and a naive standard error would overstate precision. An empty deliverable scores zero and stays in the denominator. The figure script adds no statistics at all. It reads means and standard errors from the scored summary and renders them.

No number in the pipeline is hand-typed, with two qualifications that keep that honest: the

radar’s axis set and the archetype-to-rung map are literals in code, prespecified structure rather than measured values; and the 95% multiplier is 1.959964 in the scorer and 1.96 in the renderer, so a rendered interval can differ from the scored one in the fourth decimal.

6 Partial pilot record: provenance, not results

The prespecified pilot was roughly 5,900 system runs across three seeds. It completed one condition. We release what exists, because an interrupted study that quietly drops its partial rows is indistinguishable from one that dropped the inconvenient ones.

Everything below is seed 1 only; no seed-2 or seed-3 system rows exist.

Table 5: Partial pilot rows, released as provenance. Only the memoryless floor is complete and judged; the others stop where the worker became unreachable, and no row here supports a comparative claim.

system	base rows	base non-empty	twin rows	twin non-empty	judged
nomemory (floor)	125	125	125	125	250 verdicts
fulltranscript	125	125	125	109	no
grep	125	125	125	46	no
rag-lexical	125	125	125	29	no
rag	125	0	113	34	not manifested

Only the memoryless floor is complete and judged, and it is the one result this paper reports (§4.5). The other three were manifested but never judged, and the RAG baseline never reached a manifest at all, because the scorer refuses to build one with a missing twin rather than score a partial pair.

The empty deliverables are a single uniform failure, and it is not an adapter failure. Deduplicated, the errors read `codex worker failed after 2 attempts: no output.md produced`. The retrieval side was working while this happened: the RAG baseline’s base run embedded 179 times, retrieved 995 passages and assembled 721,655 characters of context, and the worker then produced nothing on all 125. This is the deprecation of §4.6 arriving as a wall of null outputs rather than as an announcement. Runs stalled 2026-09-04; the rows were committed 2026-09-14, so that commit dates the release of the record rather than the runs.

These rows carry no comparative claim. They are released as provenance for an interrupted prespecified pilot under a worker that no longer exists. Because probe validity is worker-relative (§4.2), they cannot be completed by a successor worker and they cannot be mixed with one. No ranking, no ordering, and no per-system statement is derived from this table anywhere in this paper.

7 Limitations

Everything a reader would need to discount this work is in this section, including the items that cost us the most. They are listed in rough order of how much they should change your reading.

1. **G4 failed.** Judge–human agreement is $\kappa = 0.537$ against a prespecified gate of $\kappa \geq 0.75$ (81.3% raw, $n = 150$), and every semantic verdict in this dataset inherits that error. A post-hoc cluster bootstrap added 2026-09-17 puts the 95% interval at $[0.370, 0.688]$, and its upper

bound is below the gate, so the failure does not rest on the point estimate. It is not uniform. The judge over-accepts on absence-phrased criteria (8 of 8 errors, FALSE on only 1 of 44 items) and under-accepts on scope criteria (7 of 7), which cancel for this packet’s mix of kinds and would not cancel for another. The over-accepts land 11-to-3 on the counterfactual side where pair credit compounds them, the lenient class is scored on 218 of 371 instances (58.8%), and our 20-decoy audit passed this judge at 2/41 without catching it. Contained, not fatal: no side of any instance is scored on absence criteria alone, so the floor result is unaffected, but scores on absence-heavy archetypes (A4 70.8%, A7 57.0%) should be read as upper bounds. Read the diagnosis in two halves. The degenerate pass mode is demonstrated on all 5,398 committed verdicts without human labels; whether the judge is *wrong* on specific items rests on one rater.

2. **One rater, and not an independent one.** G4 was rated by a single author, who has read seed content, a disclosed downgrade from the two-rater design originally specified. There is therefore no inter-rater κ , and judge error cannot be separated from rater error. A second independent rater is the first thing a v2 should buy, and a judge panel is not a substitute for one: the companion paper [22] reports four further blinded judges agreeing with each other far above the gate and with this rater far below it, with identical verdict vectors on the sampled absence criteria.
3. **All three released seeds fail the instance gate**, at 125, 131, and 115 against a prespecified 135, and seed 3 also fails the cluster gate at 40/54 against 45. Instances were retained and marked rather than topped up, and no failing probe was repaired. Four prespecified gates have failed in this project against one clean positive result, which is an asymmetry we report rather than manage.
4. **The evaluation this dataset was built for did not happen.** The pinned worker was withdrawn by its provider mid-pilot (§4.6), so there are no comparative system numbers here, and the partial rows in §6 are provenance rather than results. Worker billing and authentication changed during the study; the model and binary pin held until deprecation ended access entirely.
5. **Simulated organizations, not real logs.** Three screened seeds, one organization type (a two-team software company at L1–L2 scale), one worker model. The external-validity claim stops there.
6. **Nothing anchors probe difficulty to a human.** The ceiling condition is the pinned worker with the relevant facts injected, so a valid item is defined entirely relative to a model, and there is no human performance baseline anywhere on the probes. Section 4.2 reports that relativity as a finding. The separate point here is that construct validity against real organizational knowledge work is argued from the design and never measured.
7. **Event streams carry no timestamps.** `sim_time` is null throughout v1 and temporal order is positional, so no result here separates “knows the order” from “reads a date”. Screening anchors saw no timestamps either, so the condition is uniform.
8. **The judge and the human rater saw the same rules through different instruments.** The judge scored every criterion for one output together; the human scored one criterion per spreadsheet row, independently, under a no-backtracking rule. The rule text is materially identical in all three places it is maintained, which we checked, but no part of the κ gap has

been attributed between judge quality and presentation. We found this while preparing the paper and did not measure it.

9. **Judge identity is procedural, not enforced.** The pipeline records the judging model as a free-text tag; nothing in code selects a model or verifies the tag against whatever actually produced the verdicts. The guarantee that the judge was never the worker model or the criterion-author family rests on protocol discipline and dated records, not on a mechanism. Anyone rebuilding on this harness should close that gap.
10. **Scoring-rule disclosures.** The scorer-exploit audit reads FAIL as literally prespecified and PASS when re-scoped to sides carrying at least one positive-content criterion; both lines are permanent in the report, and the re-scoping is post-hoc and labelled as such. The rank-direction robustness rule ($\geq 2/3$ seeds, a disclosed downgrade from $\geq 4/5$) was prespecified for a pilot that did not run. **Twenty-eight criteria fall below the 0.7 raw-agreement threshold, and the prespecified rule to drop them was measured and then declined:** applying it moves the dataset from 371 to 341 valid instances and fails all six gates instead of three, the whole 30-instance gap being sides the drop leaves with no criteria, and 25 of the 28 rest on a single sampled judgment. We committed to the rule before running it, kept 371 after seeing the result, and publish both numbers with their dates, so a reader who thinks that was the wrong call can prefer the other one.
11. **Released-harness configurations.** Market systems run their internal LLM on Gemini where configurable, with every deviation from vendor defaults listed. Shared-store configurations do not enforce per-principal visibility and v1 scores no leakage, so the silo ablation captures the benefit of sharing without its governance cost. These describe the released harness and support no claim in this paper.
12. **Vendor right-of-reply was not triggered,** because it attaches to published vendor numbers and this paper publishes none. The procedure stays specified for any re-anchored evaluation.

8 Release

The release is built and checked by a single script, so the bundle is reproducible rather than hand-assembled and the withholding policy is enforced by code rather than by care. A policy that depends on nobody making a mistake is not a policy.

- **Ships:** the SUT-facing streams, counterfactual twins, probes with their scoring assertions, the frozen valid sets, Croissant 1.0 and RAI metadata, a CC BY 4.0 data licence (code stays MIT), a maintenance plan, checksums, and a manifest naming what was withheld and why.
- **Withheld:** the ground-truth fact ledger (746 facts across the released seeds, recording which facts are probed and which are planted distractors), probe plans, realization maps, annotated scoring streams, the generator, the two unscreened holdout seeds, and every rater key.
- **Published on purpose:** the assertions. Scoring is impossible without them, and the cost of publishing them is that the set is open-book by construction, which the withheld generator, the holdout seeds and the canary strings are the answer to.

- **Verified:** the release script fails on a leaked ledger, a missing canary, a tampered file, or any withheld filename. The redacted bundle still runs. It drives the real runner to the full frozen valid set, 125 base plus 125 twin rows on seed 1.

The dataset lives at huggingface.co/datasets/notmehul/memory-bench under CC BY 4.0, and the harness, the screening pipeline and every script that produced a number here under MIT. At the time of writing the repository is private and opens with the paper; its bundle is the one this section describes, and its checksums match a byte-for-byte round trip through the host.

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Author contributions

M.S. designed the benchmark, built the generator, screening and scoring pipelines, ran every measurement, and wrote the paper. The G4 calibration packet was rated by M.S. alone, which is the limitation disclosure 2 records and the one this work most needs repaired.

Competing interests and funding

This is an independent study with no institutional affiliation and no funding from any vendor whose system appears in the inclusion criteria. No author-affiliated memory system is evaluated anywhere in this work, and no system was given advance access to the dataset or the screening pipeline. Every exclusion from the candidate set is reported with its reason.

Ethics

The organizations, people and events in this dataset are synthetic; no real personal data was collected or processed. The only human-subject component is the author’s own rating of the calibration packet.

Data and code availability

The dataset is at huggingface.co/datasets/notmehul/memory-bench under CC BY 4.0, private until the all-in-one release; the harness is MIT. Section 8 lists what ships and what is withheld, and the withholding is enforced by `scripts/release.py verify` rather than by care.

Reproducibility

Every number in this paper derives from a committed artifact, and `tests/test_paper_numbers.py` asserts the mapping. One limit is structural rather than procedural: the pinned worker was withdrawn by its provider on 2026-09-04 (§4.6), so the screening anchors cannot be reproduced under the model that produced them. The re-anchoring procedure is specified in §4.6 and ships with the release.

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